

Appendix B: Table of Integrals

Basic Integrals

1. $\int u^n du = \frac{u^{n+1}}{n+1} + C, \quad n \neq -1$
2. $\int \frac{du}{u} = \ln|u| + C$
3. $\int e^u du = e^u + C$
4. $\int a^u du = \frac{a^u}{\ln a} + C$
5. $\int \sin u du = -\cos u + C$
6. $\int \cos u du = \sin u + C$
7. $\int \sec^2 u du = \tan u + C$
8. $\int \csc^2 u du = -\cot u + C$
9. $\int \sec u \tan u du = \sec u + C$
10. $\int \csc u \cot u du = -\csc u + C$
11. $\int \tan u du = \ln|\sec u| + C$
12. $\int \cot u du = \ln|\sin u| + C$
13. $\int \sec u du = \ln|\sec u + \tan u| + C$
14. $\int \csc u du = \ln|\csc u - \cot u| + C$
15. $\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1}\left(\frac{u}{a}\right) + C$
16. $\int \frac{du}{a^2 + u^2} = \frac{1}{a} \tan^{-1}\left(\frac{u}{a}\right) + C$
17. $\int \frac{du}{u\sqrt{u^2 - a^2}} = \frac{1}{a} \sec^{-1}\left|\frac{|u|}{a}\right| + C$

Trigonometric Integrals

18. $\int \sin^2 u du = \frac{1}{2}u - \frac{1}{4}\sin 2u + C$
19. $\int \cos^2 u du = \frac{1}{2}u + \frac{1}{4}\sin 2u + C$
20. $\int \tan^2 u du = \tan u - u + C$
21. $\int \cot^2 u du = -\cot u - u + C$
22. $\int \sin^3 u du = -\frac{1}{3}(2 + \sin^2 u)\cos u + C$
23. $\int \cos^3 u du = \frac{1}{3}(2 + \cos^2 u)\sin u + C$
24. $\int \tan^3 u du = \frac{1}{2}\tan^2 u + \ln|\cos u| + C$
25. $\int \cot^3 u du = -\frac{1}{2}\cot^2 u - \ln|\sin u| + C$

52. $\int \coth u du = \ln|\sinh u| + C$
53. $\int \operatorname{sech} u du = \tan^{-1}|\sinh u| + C$
54. $\int \cosh u du = \ln\left|\tanh \frac{1}{2}u\right| + C$
55. $\int \operatorname{sech}^2 u du = \tanh u + C$
56. $\int \operatorname{csch}^2 u du = -\coth u + C$
57. $\int \operatorname{sech} u \tanh u du = -\operatorname{sech} u + C$
58. $\int \operatorname{csch} u \coth u du = -\operatorname{csch} u + C$

Inverse Trigonometric Integrals

59. $\int \sin^{-1} u du = u \sin^{-1} u + \sqrt{1 - u^2} + C$
60. $\int \cos^{-1} u du = u \cos^{-1} u - \sqrt{1 - u^2} + C$
61. $\int \tan^{-1} u du = u \tan^{-1} u - \frac{1}{2}\ln(1 + u^2) + C$
62. $\int u \sin^{-1} u du = \frac{2u^2 - 1}{4} \sin^{-1} u + \frac{u\sqrt{1 - u^2}}{4} + C$
63. $\int u \cos^{-1} u du = \frac{2u^2 - 1}{4} \cos^{-1} u - \frac{u\sqrt{1 - u^2}}{4} + C$
64. $\int u \tan^{-1} u du = \frac{u^2 + 1}{2} \tan^{-1} u - \frac{u}{2} + C$
65. $\int u^n \sin^{-1} u du = \frac{1}{n+1} \left[u^{n+1} \sin^{-1} u - \int \frac{u^{n+1} du}{\sqrt{1 - u^2}} \right], \quad n \neq -1$
66. $\int u^n \cos^{-1} u du = \frac{1}{n+1} \left[u^{n+1} \cos^{-1} u + \int \frac{u^{n+1} du}{\sqrt{1 - u^2}} \right], \quad n \neq -1$
67. $\int u^n \tan^{-1} u du = \frac{1}{n+1} \left[u^{n+1} \tan^{-1} u - \int \frac{u^{n+1} du}{1 + u^2} \right], \quad n \neq -1$

Integrals Involving $a^2 + u^2$, $a > 0$

68. $\int \sqrt{a^2 + u^2} du = \frac{u}{2}\sqrt{a^2 + u^2} + \frac{a^2}{2}\ln\left(u + \sqrt{a^2 + u^2}\right) + C$
69. $\int u^2 \sqrt{a^2 + u^2} du = \frac{u}{8}(u^2 + 2u^2)\sqrt{a^2 + u^2} - \frac{a^4}{8}\ln\left(u + \sqrt{a^2 + u^2}\right) + C$
70. $\int \frac{\sqrt{a^2 + u^2}}{u} du = \sqrt{a^2 + u^2} - a \ln\left|\frac{a + \sqrt{a^2 + u^2}}{u}\right| + C$
71. $\int \frac{\sqrt{a^2 + u^2}}{u^3} du = -\frac{\sqrt{a^2 + u^2}}{u} + \ln\left(u + \sqrt{a^2 + u^2}\right) + C$
72. $\int \frac{du}{\sqrt{a^2 + u^2}} = \ln\left(u + \sqrt{a^2 + u^2}\right) + C$
73. $\int \frac{u^3}{\sqrt{a^2 + u^2}} du = \frac{u}{2}\left(\sqrt{a^2 + u^2}\right) - \frac{a^2}{2}\ln\left(u + \sqrt{a^2 + u^2}\right) + C$
74. $\int \frac{du}{u\sqrt{a^2 + u^2}} = \frac{-1}{a} \ln\left|\frac{\sqrt{a^2 + u^2} + a}{u}\right| + C$
75. $\int \frac{du}{u^3 \sqrt{a^2 + u^2}} = -\frac{\sqrt{a^2 + u^2}}{a^2 u} + C$
76. $\int \frac{du}{(a^2 + u^2)^{3/2}} = \frac{u}{a^2 \sqrt{a^2 + u^2}} + C$

26. $\int \sec^3 u du = \frac{1}{2}\sec u \tan u + \frac{1}{2}\ln|\sec u + \tan u| + C$
27. $\int \csc^3 u du = -\frac{1}{2}\csc u \cot u + \frac{1}{2}\ln|\csc u - \cot u| + C$
28. $\int \sin^n u du = \frac{-1}{n}\sin^{n-1} u \cos u + \frac{n-1}{n}\int \sin^{n-2} u du$
29. $\int \cos^n u du = \frac{1}{n}\cos^{n-1} u \sin u + \frac{n-1}{n}\int \cos^{n-2} u du$
30. $\int \tan^n u du = \frac{1}{n-1}\tan^{n-1} u - \int \tan^{n-2} u du$
31. $\int \cot^n u du = \frac{-1}{n-1}\cot^{n-1} u - \int \cot^{n-2} u du$
32. $\int \sec^n u du = \frac{1}{n-1}\tan u \sec^{n-2} u + \frac{n-2}{n-1}\int \sec^{n-2} u du$
33. $\int \csc^n u du = \frac{-1}{n-1}\cot u \csc^{n-2} u + \frac{n-2}{n-1}\int \csc^{n-2} u du$
34. $\int \sin au \sin bu du = \frac{\sin(a-b)u}{2(a-b)} - \frac{\sin(a+b)u}{2(a+b)} + C$
35. $\int \cos au \cos bu du = \frac{\sin(a-b)u}{2(a-b)} + \frac{\sin(a+b)u}{2(a+b)} + C$
36. $\int \sin au \cos bu du = -\frac{\cos(a-b)u}{2(a-b)} - \frac{\cos(a+b)u}{2(a+b)} + C$
37. $\int u \sin u du = \sin u - u \cos u + C$
38. $\int u \cos u du = \cos u + u \sin u + C$
39. $\int u^n \sin u du = -u^n \cos u + n \int u^{n-1} \cos u du$
40. $\int u^n \cos u du = u^n \sin u - n \int u^{n-1} \sin u du$

Exponential and Logarithmic Integrals

42. $\int u e^{au} du = \frac{1}{a^2}(au - 1)e^{au} + C$
43. $\int u^n e^{au} du = \frac{1}{a}u^n e^{au} - \frac{n}{a} \int u^{n-1} e^{au} du$
44. $\int e^{au} \sin bu du = \frac{e^{au}}{a^2 + b^2}(a \sin bu - b \cos bu) + C$
45. $\int e^{au} \cos bu du = \frac{e^{au}}{a^2 + b^2}(a \cos bu + b \sin bu) + C$
46. $\int \ln u du = u \ln u - u + C$
47. $\int u^n \ln u du = \frac{u^{n+1}}{(n+1)^2}[(n+1)\ln u - 1] + C$
48. $\int \frac{1}{u \ln u} du = \ln|\ln u| + C$

Hyperbolic Integrals

49. $\int \sinh u du = \cosh u + C$
50. $\int \cosh u du = \sinh u + C$
51. $\int \tanh u du = \ln \cosh u + C$

Integrals Involving $u^2 - a^2$, $a > 0$

77. $\int \sqrt{u^2 - a^2} du = \frac{u}{2}\sqrt{u^2 - a^2} - \frac{a^2}{2}\ln|u + \sqrt{u^2 - a^2}| + C$
78. $\int u^2 \sqrt{u^2 - a^2} du = \frac{u}{8}(2u^2 - a^2)\sqrt{u^2 - a^2} + \frac{a^4}{8}\ln|u + \sqrt{u^2 - a^2}| + C$
79. $\int \frac{\sqrt{u^2 - a^2}}{u} du = \sqrt{u^2 - a^2} - a \cos^{-1}\frac{a}{|u|} + C$
80. $\int \frac{\sqrt{u^2 - a^2}}{u^3} du = -\frac{\sqrt{u^2 - a^2}}{u} + \ln|u + \sqrt{u^2 - a^2}| + C$
81. $\int \frac{du}{\sqrt{u^2 - a^2}} = \ln|u + \sqrt{u^2 - a^2}| + C$
82. $\int \frac{u^3}{\sqrt{u^2 - a^2}} du = \frac{u}{2}\sqrt{u^2 - a^2} + \frac{a^2}{2}\ln|u + \sqrt{u^2 - a^2}| + C$
83. $\int \frac{du}{u^2 \sqrt{u^2 - a^2}} = \frac{\sqrt{u^2 - a^2}}{a^2 u} + C$
84. $\int \frac{du}{(u^2 - a^2)^{3/2}} = -\frac{u}{a^2 \sqrt{u^2 - a^2}} + C$

Integrals Involving $a^2 - u^2$, $a > 0$

85. $\int \sqrt{a^2 - u^2} du = \frac{u}{2}\sqrt{a^2 - u^2} + \frac{a^2}{2}\sin^{-1}\frac{u}{a} + C$
86. $\int u^2 \sqrt{a^2 - u^2} du = \frac{u}{8}(2u^2 - a^2)\sqrt{a^2 - u^2} + \frac{a^4}{8}\sin^{-1}\frac{u}{a} + C$
87. $\int \frac{\sqrt{a^2 - u^2}}{u} du = \sqrt{a^2 - u^2} - a \ln\left|\frac{a + \sqrt{a^2 - u^2}}{u}\right| + C$
88. $\int \frac{\sqrt{a^2 - u^2}}{u^3} du = \frac{-1}{u}\sqrt{a^2 - u^2} - \sin^{-1}\frac{u}{a} + C$
89. $\int \frac{u^3}{\sqrt{a^2 - u^2}} du = \frac{1}{2}\left(-u\sqrt{a^2 - u^2} + a^2 \sin^{-1}\frac{u}{a}\right) + C$
90. $\int \frac{du}{u\sqrt{a^2 - u^2}} = -\frac{1}{a} \ln\left|\frac{a + \sqrt{a^2 - u^2}}{u}\right| + C$
91. $\int \frac{du}{u^3 \sqrt{a^2 - u^2}} = -\frac{1}{a^2 u}\sqrt{a^2 - u^2} + C$
92. $\int (a^2 - u^2)^{3/2} du = -\frac{u}{8}(2u^2 - 5a^2)\sqrt{a^2 - u^2} + \frac{3a^4}{8}\sin^{-1}\frac{u}{a} + C$
93. $\int \frac{du}{(a^2 - u^2)^{3/2}} = -\frac{u}{a^2 \sqrt{a^2 - u^2}} + C$

Integrals Involving $2au - u^2$, $a > 0$

94. $\int \sqrt{2au - u^2} du = \frac{u-a}{2}\sqrt{2au - u^2} + \frac{a^2}{2}\cos^{-1}\left(\frac{a-u}{a}\right) + C$
95. $\int \frac{du}{\sqrt{2au - u^2}} = \cos^{-1}\left(\frac{a-u}{a}\right) + C$
96. $\int u\sqrt{2au - u^2} du = \frac{2u^2 - au - 3a^2}{6}\sqrt{2au - u^2} + \frac{a^3}{2}\cos^{-1}\left(\frac{a-u}{a}\right) + C$
97. $\int \frac{du}{u\sqrt{2au - u^2}} = -\frac{\sqrt{2au - u^2}}{au} + C$

Integrals Involving $a + bu$, $a \neq 0$

98. $\int \frac{u}{a + bu} du = \frac{1}{b^2}(a + bu - a \ln|a + bu|) + C$

$$99. \int \frac{u^2}{a+bu} du = \frac{1}{2b^3} [(a+bu)^2 - 4a(a+bu) + 2a^2 \ln|a+bu|] + C$$

$$100. \int \frac{du}{u(a+bu)} = \frac{1}{a} \ln \left| \frac{u}{a+bu} \right| + C$$

$$101. \int \frac{du}{u^2(a+bu)} = -\frac{1}{au} + \frac{b}{a^2} \ln \left| \frac{a+bu}{u} \right| + C$$

$$102. \int \frac{u}{(a+bu)^2} du = \frac{a}{b^2(a+bu)} + \frac{1}{b^2} \ln|a+bu| + C$$

$$103. \int \frac{u}{u(a+bu)^2} du = \frac{1}{a(a+bu)} - \frac{1}{a^2} \ln \left| \frac{a+bu}{u} \right| + C$$

$$104. \int \frac{u^3}{(a+bu)^2} du = \frac{1}{b^3} \left(a+bu - \frac{a^2}{a+bu} - 2a \ln|a+bu| \right) + C$$

$$105. \int u\sqrt{a+bu} du = \frac{2}{15b^3} (3bu-2a)(a+bu)^{3/2} + C$$

$$106. \int \frac{u}{\sqrt{a+bu}} du = \frac{2}{3b^2} (bu-2a)\sqrt{a+bu} + C$$

$$107. \int \frac{u^2}{\sqrt{a+bu}} du = \frac{2}{15b^3} (8a^2+3b^2u^2-4abu)\sqrt{a+bu} + C$$

$$108. \int \frac{du}{u\sqrt{a+bu}} = \begin{cases} \frac{1}{\sqrt{a}} \ln \left| \frac{\sqrt{a+bu}-\sqrt{a}}{\sqrt{a+bu}+\sqrt{a}} \right| + C, & \text{if } a > 0 \\ \frac{a}{\sqrt{-a}} \tan^{-1} \sqrt{\frac{a+bu}{-a}} + C, & \text{if } a < 0 \end{cases}$$

$$109. \int \frac{\sqrt{a+bu}}{u} du = 2\sqrt{a+bu} + a \int \frac{du}{u\sqrt{a+bu}}$$

$$110. \int \frac{\sqrt{a+bu}}{u^2} du = -\frac{\sqrt{a+bu}}{u} + \frac{b}{2} \int \frac{du}{u\sqrt{a+bu}}$$

$$111. \int u^n \sqrt{a+bu} du = \frac{2}{b(2n+3)} \left[u^n(a+bu)^{3/2} - na \int u^{n-1} \sqrt{a+bu} du \right]$$

$$112. \int \frac{u^n}{\sqrt{a+bu}} du = \frac{2u^n \sqrt{a+bu}}{b(2n+1)} - \frac{2na}{b(2n+1)} \int \frac{u^{n-1}}{\sqrt{a+bu}} du$$

$$113. \int \frac{du}{u^n \sqrt{a+bu}} = -\frac{\sqrt{a+bu}}{a(n-1)u^{n-1}} - \frac{b(2n-3)}{2a(n-1)} \int \frac{du}{u^{n-1} \sqrt{a+bu}}$$

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